

```

1 char input[11];
2 int count = 0;
3
4 void setup()
5 {
6   Serial.begin(9600);
7 }
8
9 void loop()
10 {
11   if(Serial.available())
12   {
13     count = 0;
14     while(Serial.available() && count < 11)
15     {
16       input[count] = Serial.read();
17       count++;
18       delay(5);
19     }
20     for(int i=0;i<11;i++)
21     {
22       Serial.print(input[i]);
23     }
24   }

```

Fig. 3: Snippet of source code

number generators (PRNG) and some simple cryptographic operations.

Since the current generation of tags comes with built-in pseudo-random generators, implementing these operations is possible with low complexity. The secret information stored inside the tags is communicated in a more secure way, ensuring confidentiality, integrity, and authentication. The security of our proposed scheme is analysed against different attacks on RFID and compared with the performance of some existing protocols. Experimental results demonstrate a significant improvement in security with an average cost when compared with existing techniques.

This is a proposed design of a smart and secure system topology for login using RFID technology. With this technology, the device can be configured to authenticate and then set selected functions, such as a login system or attendance system.

The EM 18 RFID module is used to read the RFID tags and devices, and Arduino is used to process the data and set functions like lock, unlock, or authentication. If the correct RFID tag is scanned,

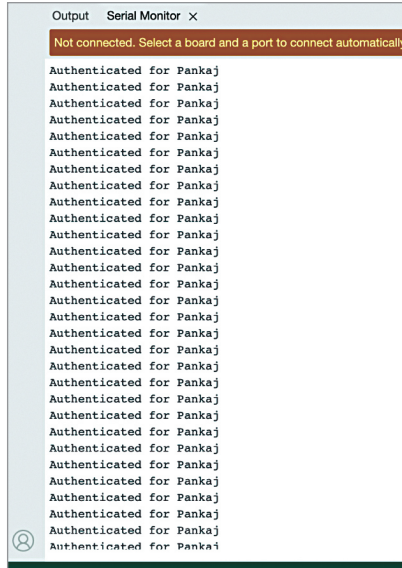


Fig. 4: Snippet of serial port

it authenticates and recognises the person, displaying the result in the serial monitor. Otherwise, it indicates that the RFID is not correct.

Fig. 1 shows the prototype of the project tested at EFY Lab. The components used in the project are listed in the Bill of Materials table.

Circuit diagram

Fig. 2 shows the circuit diagram of the project, which is built around an Arduino board (MOD1) and EM18 RFID reader module (MOD2). Connect the RFID module as shown in Fig. 2. Here, the RFID module uses the same RX pin (D0) of the Arduino that is used for hardware

code upload. So, if you have connected that pin and then uploaded the code, you will get the uploading error. While uploading code, make sure you disconnect the TX pin of the EM 18 module from the D0 pin of Arduino board. After uploading the code, open the serial monitor and connect the serial pins of the EM 18 module to Arduino pins.

Software

The code is developed using the Arduino IDE. The EM 18 RFID module operates on serial communication; when RFID is detected, it sends the data over serial pins. Here, the Arduino hardware serial pin RX (D0) has been used. In the code, the setup function is created, and the serial port is started using the specified baud rate of EM 18, which is generally 9600. Then, the loop function is created, and the serial data availability is checked. If the right RFID is detected, it prints in the serial monitor along with the name.

Upload the source code by selecting the port and board. Next, change the RFID number to the correct RFID number of your card in the code and upload the code. Fig. 3 shows a snippet of the source code.

Testing

To test, connect the Arduino to a laptop, open the serial port, and then bring the right RFID tag near the EM 18 module. If it is the correct RFID tag, it will print or perform as has been set in the code. If it detects the right RFID card, it prints on the serial monitor: 'authenticated for Pankaj' (author's name), as shown in the snippet of the serial port in Fig. 4. **EFY**

EFY Note
The source code
of this project
is available

for free download at
www.electronicsforu.com



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